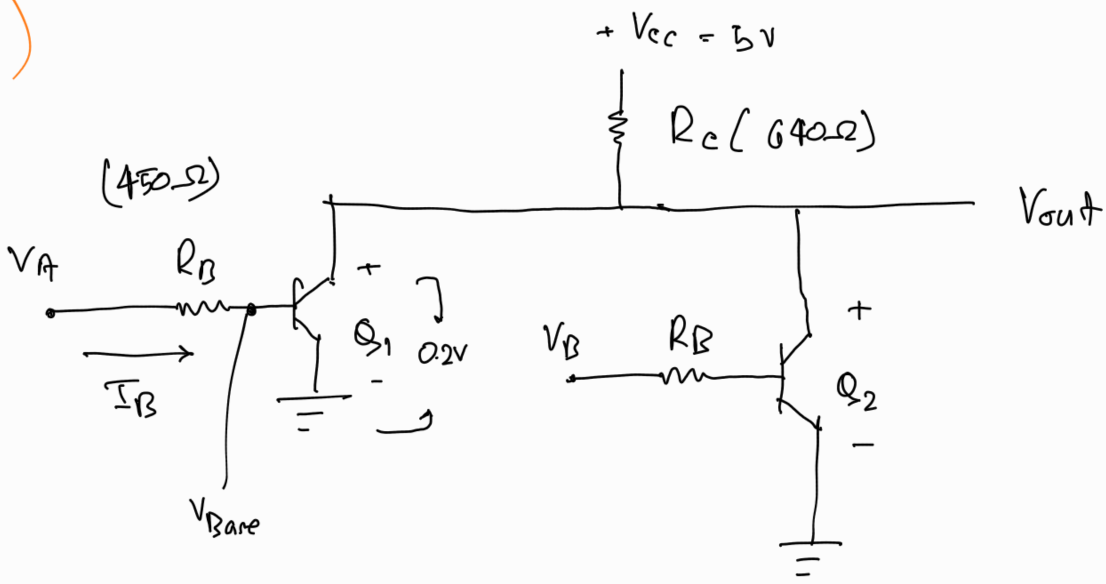
23/07/25

(class-e!)

NOR gate using RTL



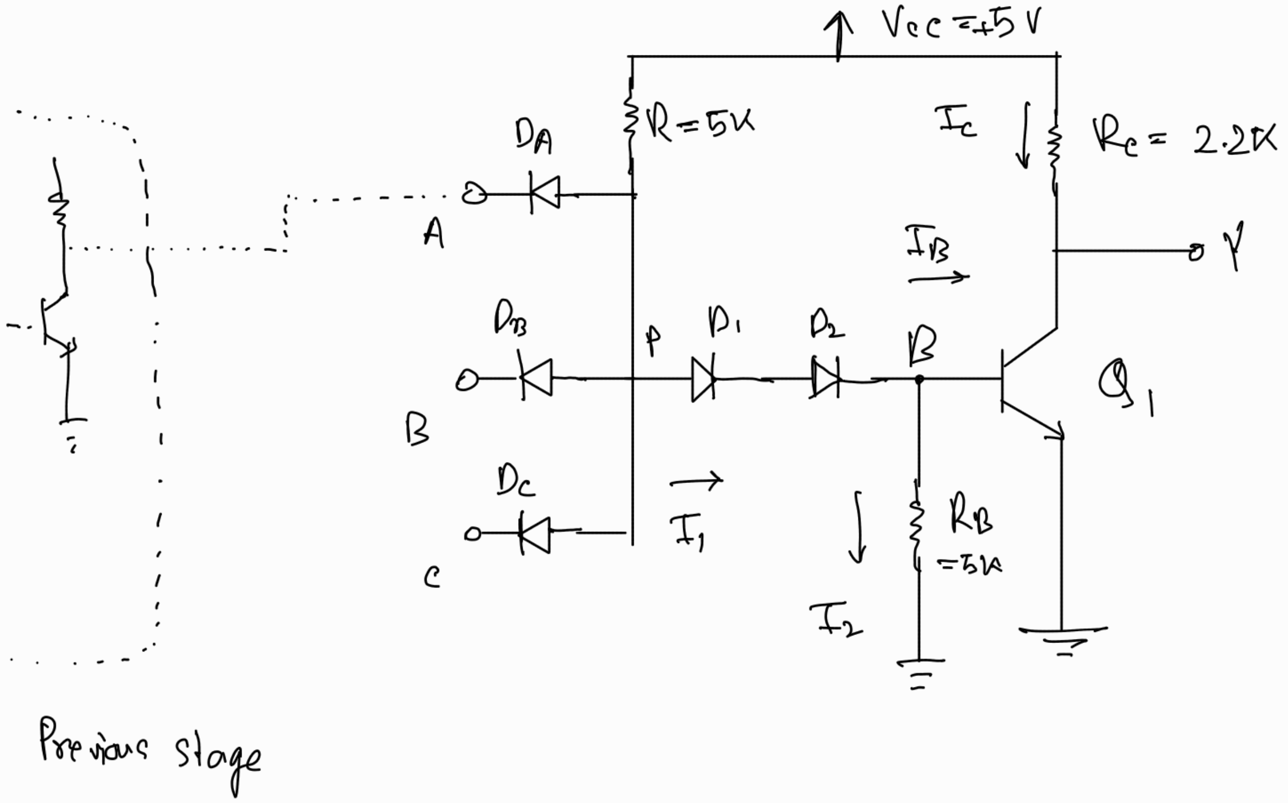
Low → 0v

High → 5v

Truth table

V_A	V_B	V_{out}
0	0	1
0	1	0
1	0	0
1	1	0

NAND Gate using DTL



Assumptions

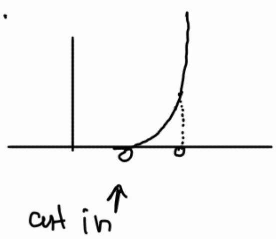
$V_{BE}(sat) = 0.8V$

$V_{BE}(cut\ in) = 0.5V$

$V_{CE}(sat) = 0.2V$

What is cut in?

The voltage at which the Base emitter junction flow just starts is called cut in.



voltage drop across a conducting diode = 0.7v

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Case - 1

$V_A \rightarrow$ low

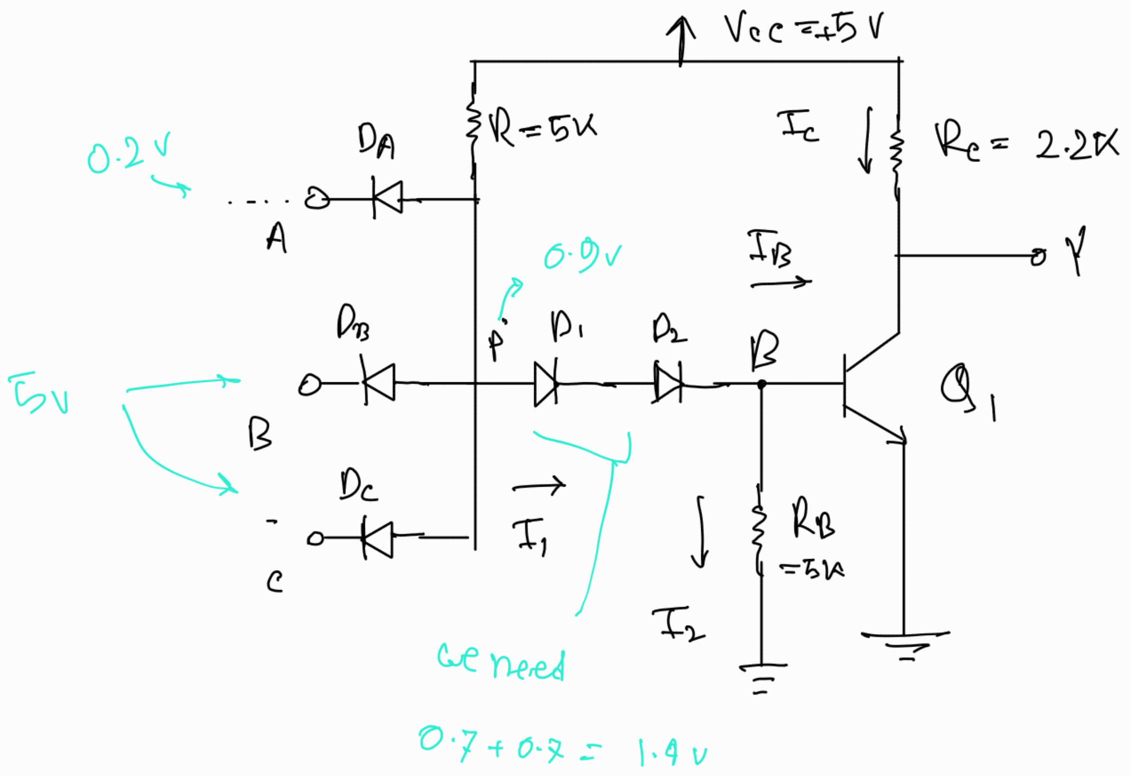
$V_B \rightarrow$ high

$V_C \rightarrow$ high

also any other permutation except 111

For logic low, gate transistor is saturated and output voltage is 0.2 V

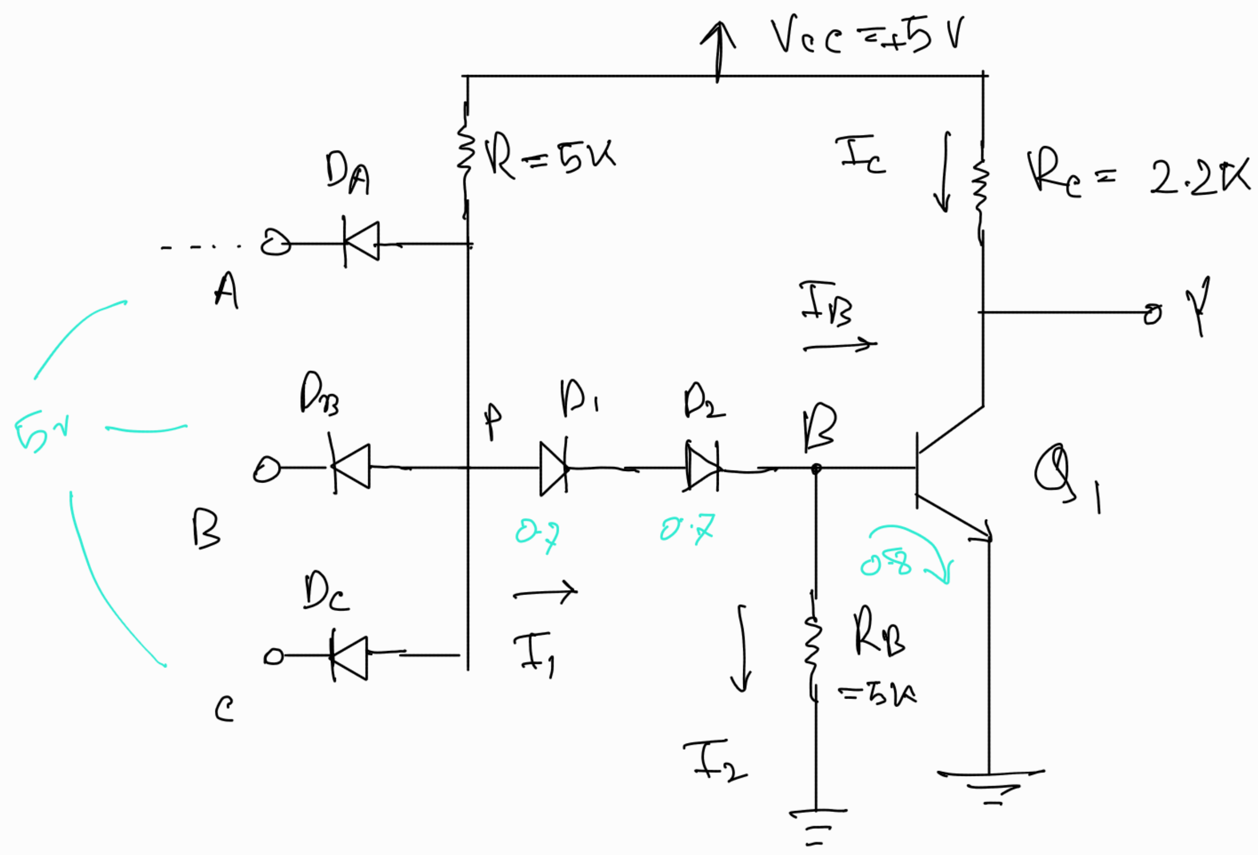
For logic high, gate transistor is turned off and output voltage is 5 V



Since V_A is logic low D_A conducts (forward bias)

$$\therefore V_P = 0.2 + 0.7 = 0.9$$

Case-2:



$0.7 + 0.7 + 0.8 = 2.2$

Value of I_1, I_2, I_B, I_C

$$I_1 = \frac{V_{cc} - V_{D}}{R} =$$

$$I_2 = \frac{V_{BE}(sat)}{R_B}$$

$$I_B = I_1 - I_2 = 0.56\,mA - 0.16\,mA$$

$$I_C = \frac{V_{cc} - V_{CE}(sat)}{R_C} = \frac{5 - 0.2}{2.2k} = 2.18\,mA$$

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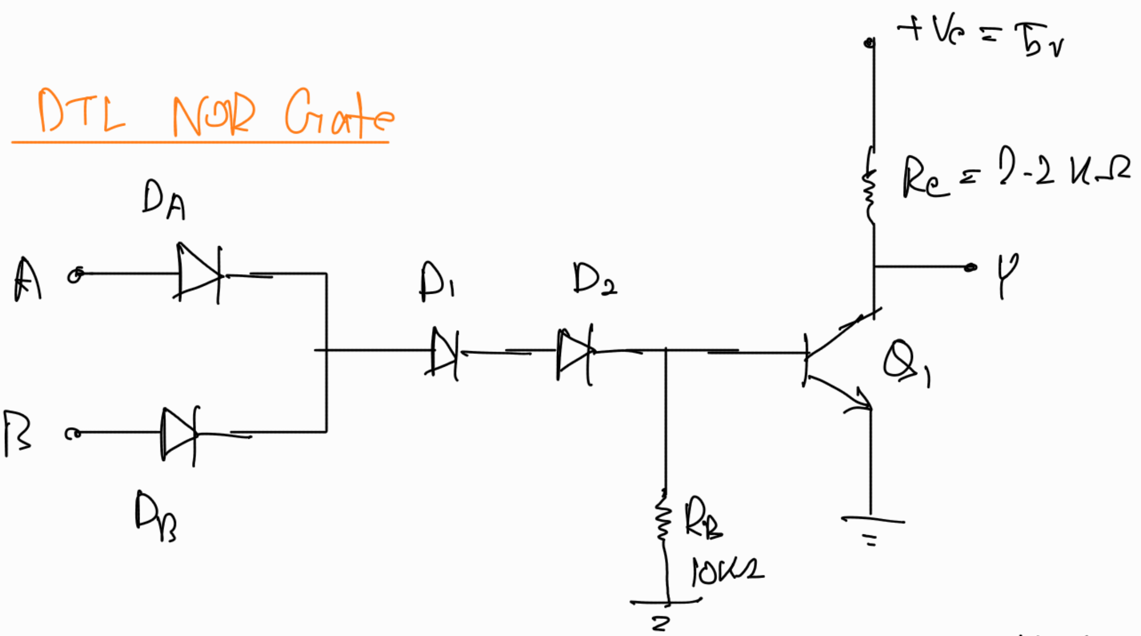
Transistor - Transistor Logic (TTL)

DTL NOR Gate



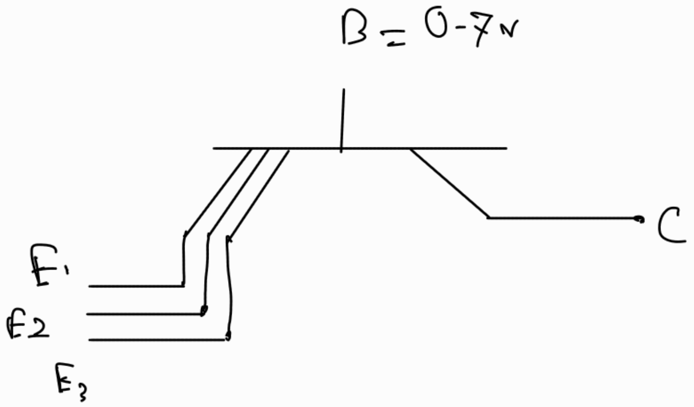
High – 5v

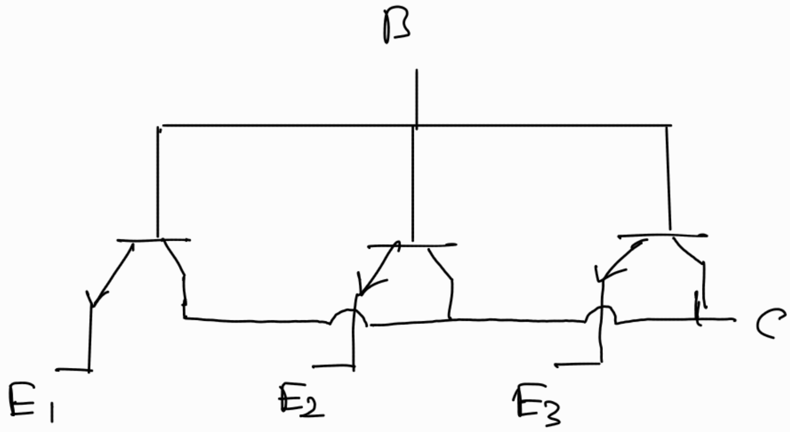
Low – 0.2v

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

$0.7 + 0.7 + 0.7 + 0.8 = 2.9$

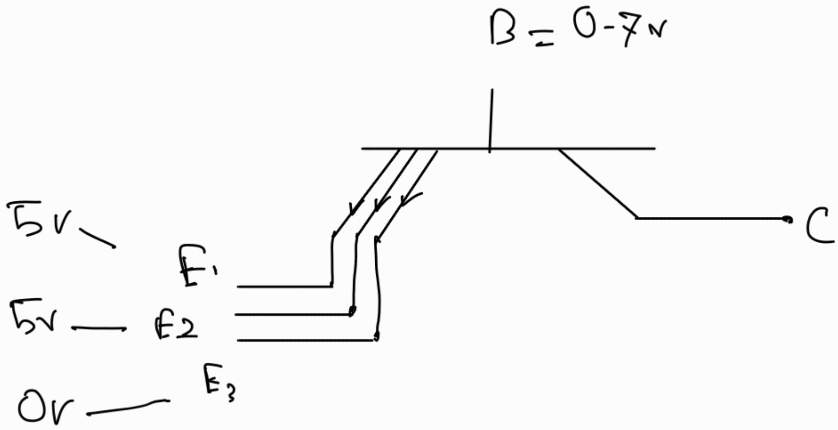
Multi-emitter transistor :





Actually looks like

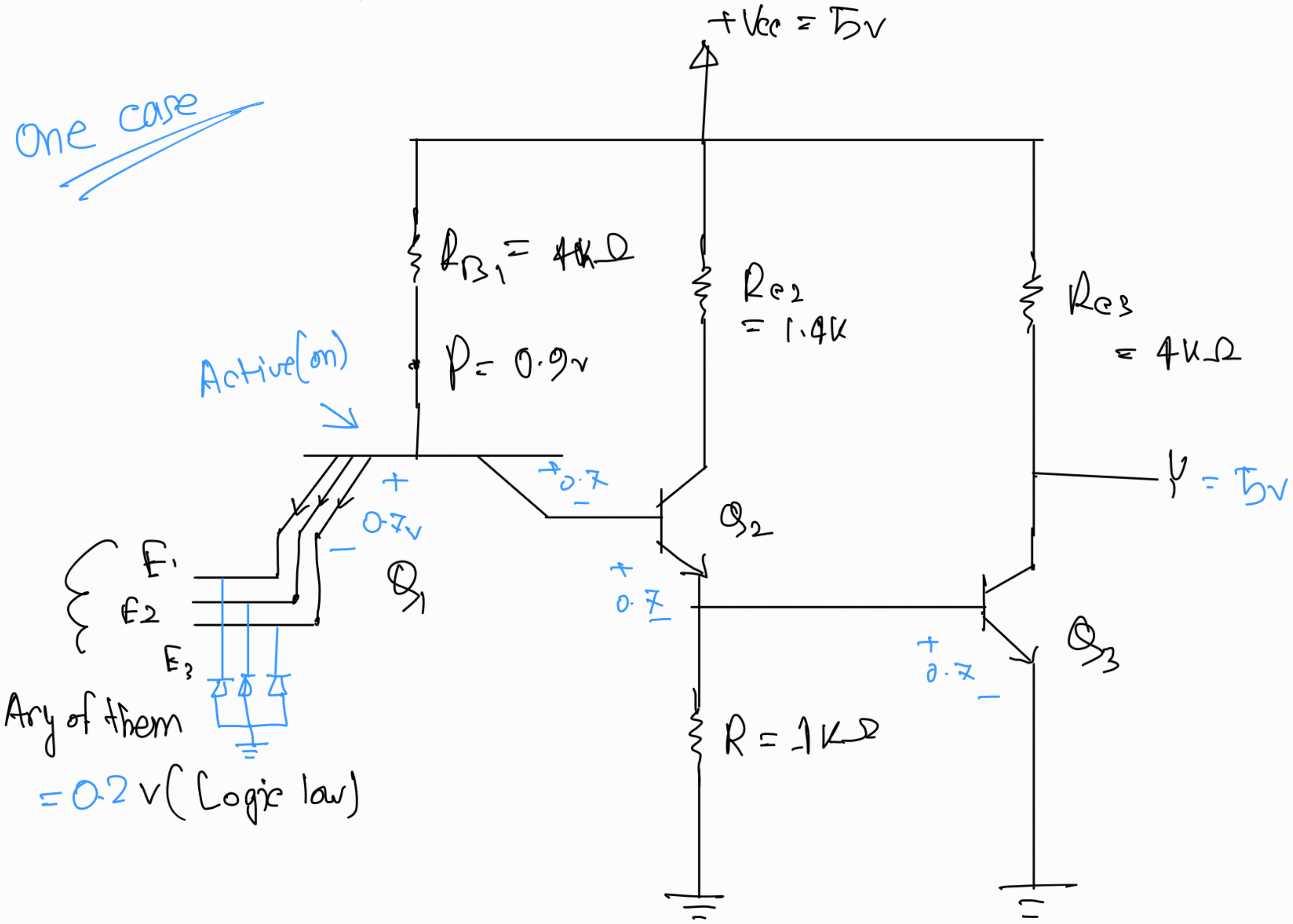
For this condition



Reverse bias

Reverse bias

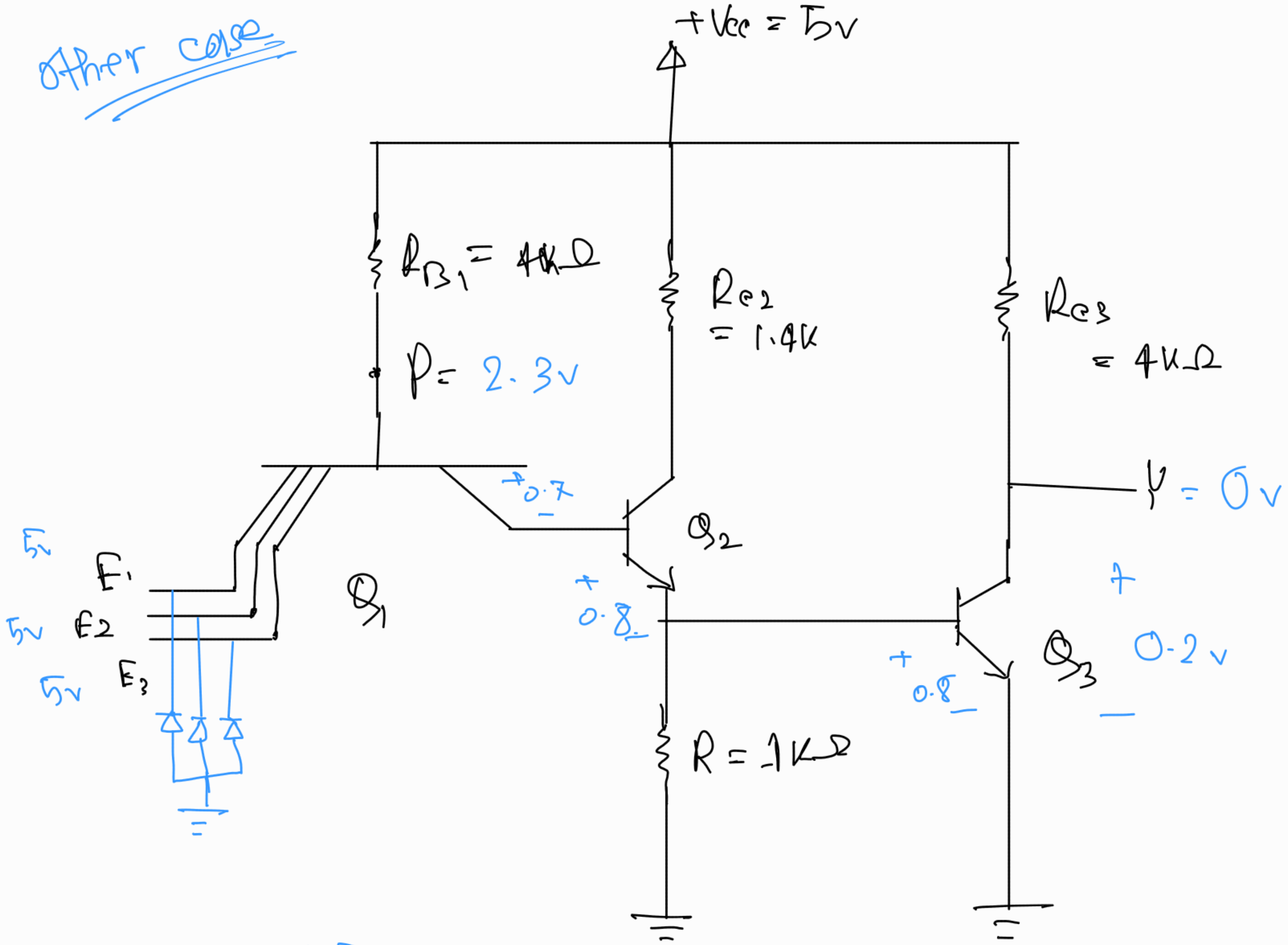
TTL NAND gate



$0.7 + 0.7 + 0.7 = 2.1\ v > P = 0.9$

So all the flow will be at Y = 5 v

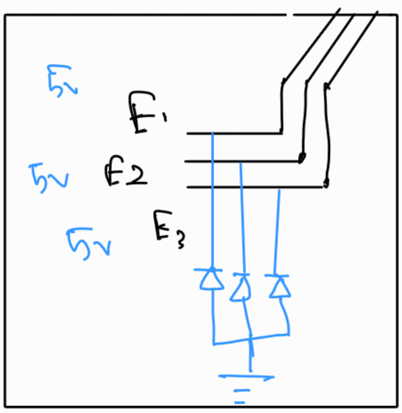
other case



$0.7 + 0.8 + 0.8$

$= 2.3v$

Time taken to discharge capacitor, T = Rc

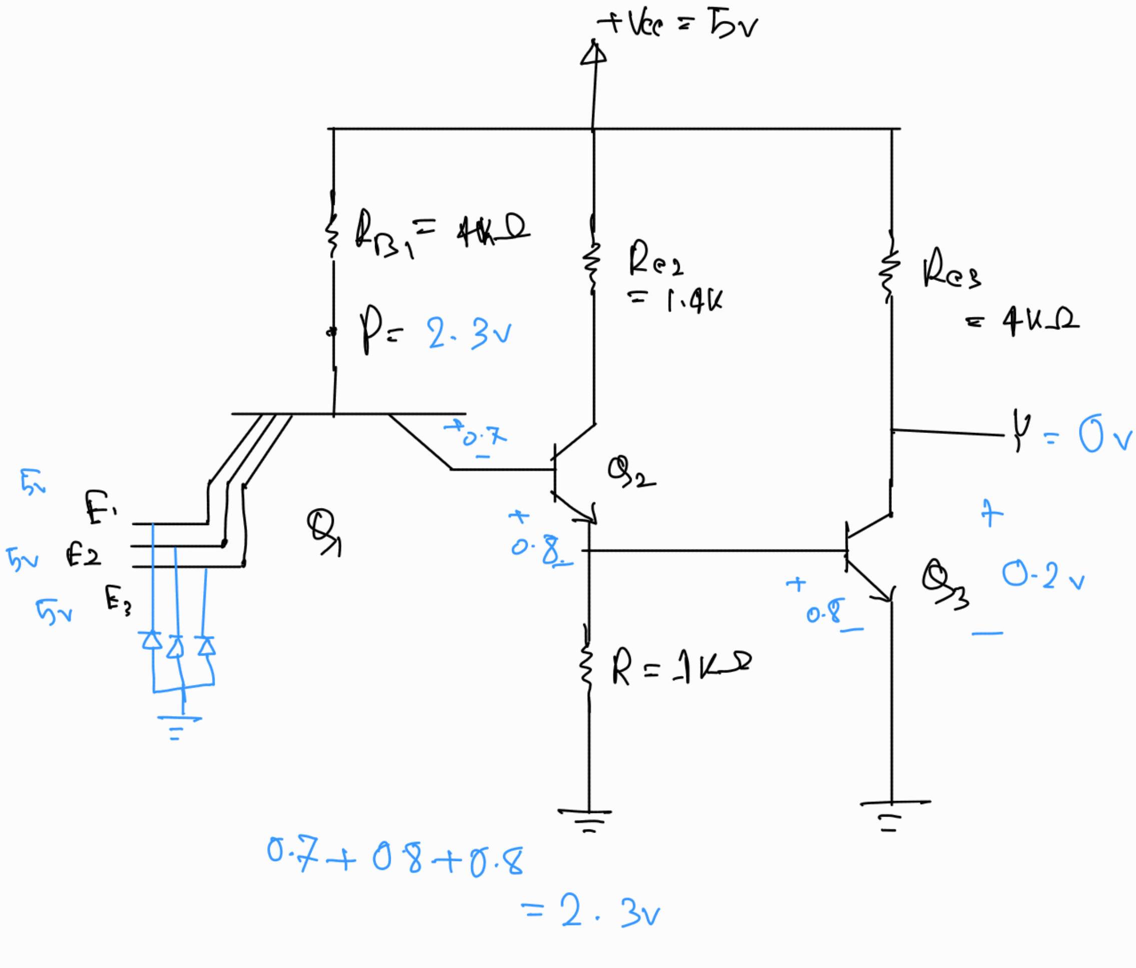


what is the purpose of 3 diodes?

They will clamp the signal if it goes below 0 reducing noise

Math:

In the circuit of TTL NAND gate, if all the inputs are in High condition then determine the emitter current of Q_3.



Ans:

$V_A = 5v$

$V_B = 5v$

$V_C = 5v$

→ Q_1 = reverse action mode

→ B-E junction → rev. Bias

→ B-C junction → fwd Bias

$$V_P = V_{BC_1} + V_{BE_2} + V_{BE_3}$$

$$= 0.7 + 0.8 + 0.8 = 2.3v$$

$\therefore I_{B_1} = \dfrac{V_{cc} - V_p}{4k\Omega} = \dfrac{5-2.3}{4k\Omega} = 0.675\,mA$

$I_{E_1} = 0$ (Reverse bias)

$0 = I_{B_1} + I_{C_1}$

$\approx I_{B_1} = -I_{C_1} = 0.675\,mA$

Here, $-I_{C_1} = I_{B_2}$

$\therefore I_{B_2} = 0.675\,mA$

$V_{BE_3} \approx V_{B_3}$ (Emitter connected to gnd) $= 0.8v$

?? I_R or $I_{1k} = \dfrac{V_{B_3}}{1k\Omega} = \dfrac{0.8}{1k} = 0.8\,mA$

$V_{C_2} = V_{B_3} + V_{CE_2} = 0.8 + 0.2 = 1v$

Now,

$I_{C_2} = \dfrac{V_{cc} - V_{C_2}}{1.4k} = \dfrac{5-1}{1.4k} = 2.86\,mA$

$I_{E_2} = I_{B_2} + I_{C_2} = (0.675 + 2.86) = 3.53\,mA$

$$I_{B3} = I_{E2} - I_{1K} = 3.53 - 0.8 = 2.73\,mA$$

$$I_{C3} = \frac{V_{CC} - 0.2}{4k} = \frac{5 - 0.2}{4k} = 1.2\,mA$$

$$I_{E3} = I_{B3} + I_{C3} = 2.73 + 1.2 = 3.93\,mA$$

(Ans)

20/07/25

TTL Improvement in output stages

Load capacitance

is the total capacitance that the output of a logic gate must charge or discharge during a logic transition (HIGH $\longleftrightarrow$ LOW)

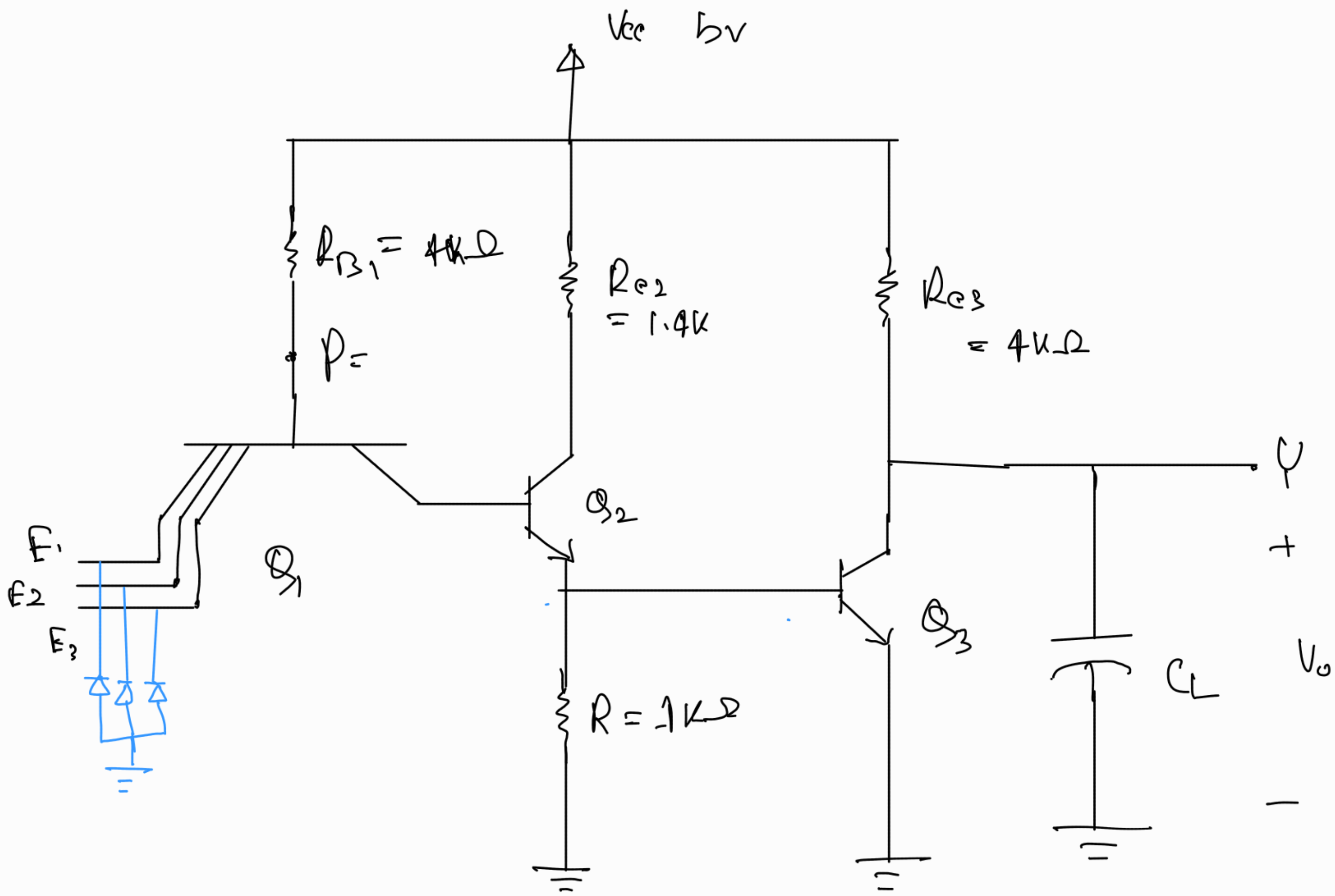
This capacitance introduces delays in signal transitions, limiting switching speed.

Sources of load capacitance:

Every TTL input pin behaves like a small capacitor due to the base-emmiter junction of the input BJT

{ in reverse bias condition → capacitor due to depletion region acting as dielectric between two charge regions }

→ The physical wires or PCB traces connecting gates act like capacitors between trace to GND, trace/signal line to adjust signal lines.



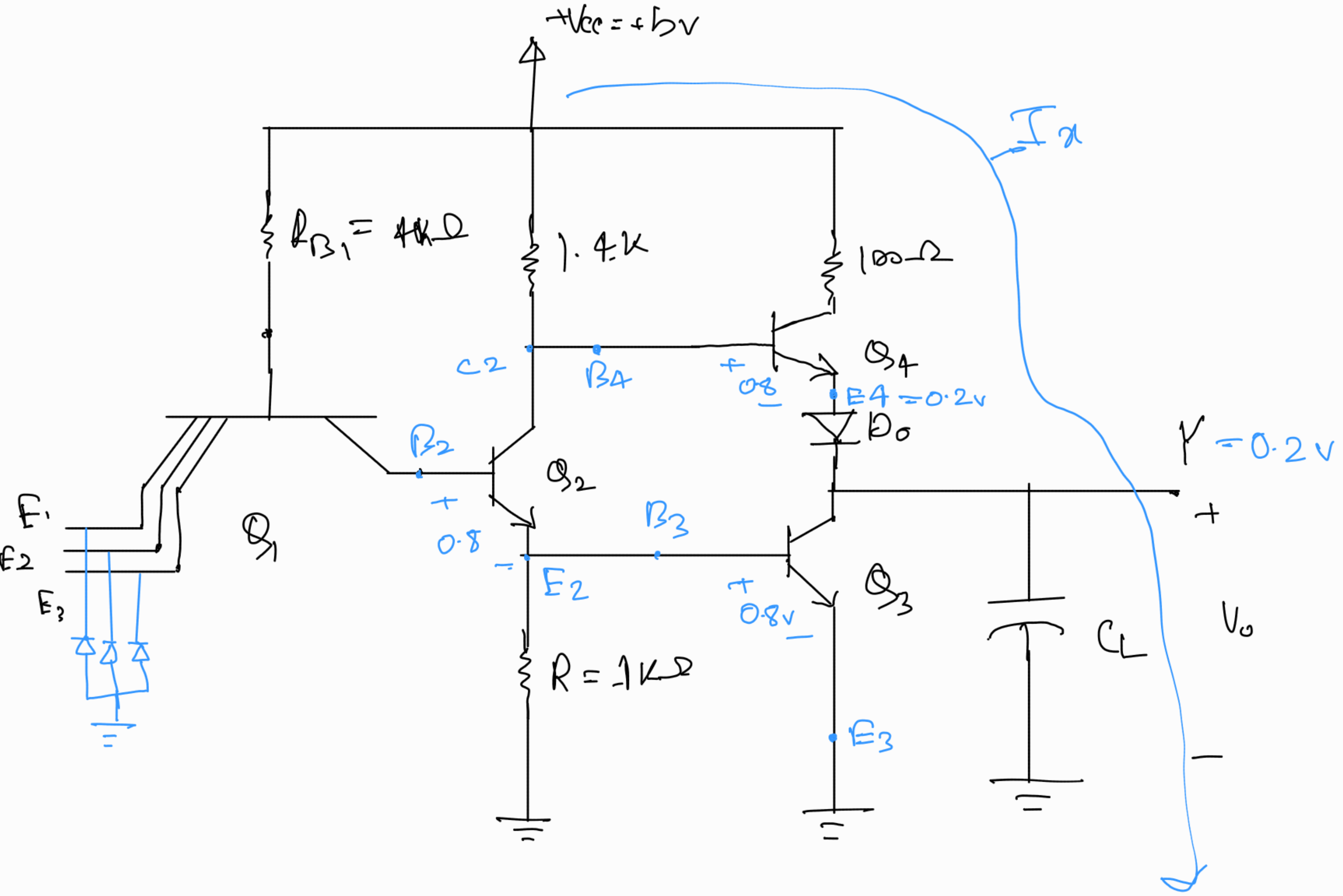
* TTL output must charge and discharge this load capacitance every time the logic level changes.

* If passive pull up resistor R_c is used for this purpose →

time constant $R_c C_L$ is high, causing long delay time when logic level experiences a transition.

~~Suddenly~~ → Shuddenly 

* Totem-pole config



$$\frac{5 - 0.2 - 0.7 - 0.4}{100} = I_x$$

$V_0 = V_{cc} - V_{BE_4}(\text{cut in}) - V_{D_0}(\text{cut in})$

$= 5 - 0.5 - 0.6$

$= 3.9v$